

Industry Relevance

The US consumer electronics market, specifically mobiles and tablets, is one of the most competitive retail segments in the country. Americans spend over \$400 billion annually on consumer electronics, and online channels account for a growing share of that. Retailers like Best Buy, Amazon, and Walmart constantly compete on pricing, promotions, and delivery experience.

The core challenge here, understanding what drives sales and forecasting demand, is something every electronics retailer faces. Getting it wrong means either overstocking (tying up capital in unsold inventory) or stockouts (losing sales to competitors).

This problem is universal, well understood by practitioners, and directly tied to profitability. Companies have publicly discussed using data analytics to optimize inventory and marketing spend. This mirrors a real operational decision that merchandising and marketing teams make on a weekly basis.

Business Context

ElectraHub is a mid-size US online retailer specializing in mobiles and tablets. They sell across four regions (North, South, East, West), with the West and East being their strongest markets, accounting for roughly 28% and 27% of transactions respectively.

Their product catalog spans a wide price range, from budget devices around \$10 to premium flagships near \$2,000, with an average product price of approximately \$1,435. They run regular promotional campaigns with discounts averaging around 29%, ranging from 0% to about 40%.

Advertising expenditure per product-campaign cycle averages around \$608, and they track campaign engagement scores that range from 0 to nearly 99. They maintain customer ratings averaging 4.4 out of 5.

Objective

The goal is to identify the key factors that drive sales performance across advertising, pricing, promotions, and customer engagement through structured univariate, bivariate, and multivariate analysis, and to build a linear regression model (simple and multiple) that quantifies the individual and combined impact of each lever on sales, enabling data-backed decisions on marketing budget allocation and inventory planning.

Problem Statement

Current State

ElectraHub's inventory planning and marketing budget allocation are largely driven by gut feel and simple historical averages. The merchandising team orders stock based on last quarter's numbers plus a blanket 10% buffer, which leads to frequent mismatches. They end up overstocking on slow-moving mid-range devices and running into stockouts on high-demand products during promotional periods.

Marketing spend (averaging about \$608 per campaign cycle) is allocated evenly across regions and product categories without any understanding of which levers actually move sales. Sales vary enormously, from as low as \$3,258 to as high as \$34,421 per product-campaign cycle, which is a 10x range. Yet the team has no model to explain or predict this variance.

They currently have no systematic way to answer basic questions like "if we increase ad spend by \$100, how much incremental sales can we expect?" or "are discounts or campaign engagement more impactful for driving sales?"

Future State

ElectraHub wants to build a regression model that can forecast sales with reasonable accuracy and, more importantly, identify the key factors (advertising spend, discounts, pricing, campaign engagement, ratings, etc.) that drive sales up or down. The goal is to move from gut-based decisions to data-informed ones, allocating marketing budgets to the highest-impact levers and setting inventory levels based on predicted demand rather than flat rules.

Context and Constraints

People

The analytics team is small. Two business analysts who are comfortable with Excel and basic dashboards but have limited experience with statistical modeling. The data science work will be done by a consultant. The merchandising and marketing managers are the end consumers of the model's output, so interpretability matters. They need to understand what the model says and why, not just see a number. This rules out black-box models and makes linear regression a natural fit because the coefficients directly tell them "a \$1 increase in ad spend is associated with \$X change in sales."

Process

Inventory planning happens quarterly with a monthly review cycle. Marketing budget allocation is decided at the start of each quarter and rarely adjusted mid-cycle. There's no feedback loop. Nobody tracks whether last quarter's spending decisions actually moved the needle. Discounting strategy is driven by competitive pressure ("competitor X is running 30% off, so we should too") rather than data on what discount level optimizes sales.

Platform

The data lives in spreadsheets and basic reporting tools. There is no ML infrastructure, no model deployment pipeline, and no real-time scoring requirement. The output of this project will be a model and a set of actionable recommendations presented to leadership, not a production API. This further supports keeping the scope to linear regression and EDA.

Past Attempts

ElectraHub has tried two approaches previously.

First, about a year ago, the marketing team ran a basic correlation analysis in Excel comparing ad spend to sales at a high level. They concluded "more spend equals more sales" but could not quantify the relationship or control for other factors like pricing and discounts happening simultaneously. The insight was too vague to act on. It did not tell them how much additional spend was worth it or whether the money would be better spent on deeper discounts instead.

Second, the merchandising team tried categorizing products into "High" and "Low" popularity buckets and stocking accordingly. But popularity alone did not capture the full picture. A "High" popularity product with a low discount in a weak region could still underperform, leading to overstock.

Both attempts failed because they looked at one variable at a time. Neither approach controlled for confounding factors or quantified the relative importance of each lever, which is exactly what a multiple linear regression model is designed to do.

Proposed Solution

The plan is to build a linear regression model (simple and multiple) to forecast sales and identify the key factors influencing sales performance. The approach follows an AI-assisted, human-in-the-loop workflow where each step is prompted, reviewed, and agreed upon before execution.

- Project setup and business question formulation

- Data understanding and data cleaning (with a prompt, plan, agree, then execute workflow)
- Feature engineering only where business questions demand it
- EDA: univariate, bivariate, multivariate, all grounded to business questions
- Encoding categorical variables and train-test split
- Simple linear regression (one predictor at a time)
- Multiple linear regression (all significant predictors combined)
- Coefficient interpretation and evaluation metrics (R^2 , Adjusted R^2 , RMSE, MAE)
- Assumption checks via plots (residual plots, scatter plots, Q-Q plot, correlation heatmap)
- Business recommendations

Solution Architecture

This follows a structured, step-by-step AI-assisted workflow where the human reviews, agrees, and drives decisions at each stage.

Step 1: Project Setup

Create the ML project folder structure with folders for data (raw, processed), notebooks (EDA, modeling), outputs (plots, reports), and a `business_questions.txt` file. Place the raw `Sales.csv` in the data folder. Document the business context and objective in a reference file so every subsequent step is grounded.

Step 2: Business Question Formulation

Provide AI with the business context and objective. Ask it to suggest two sets of questions. The first set should be answerable through EDA alone, for example: "Which region contributes the most to sales?", "Is there a visible relationship between advertising expenditure and sales?", or "How are discounts distributed across popularity levels?"

The second set should be answerable only after building the regression model, for example: "What is the expected increase in sales for every \$100 increase in ad spend, holding other factors constant?", "Which factor has the strongest influence on sales?", or "Does product price have a statistically significant negative effect on sales?"

Write all of these to `business_questions.txt`. This file becomes the guiding thread for the entire project.

Step 3: Data Understanding

Provide AI with the dataset description (column names, types, what each column represents). Ask it to produce a written summary covering the number of records, number of features, target variable, feature ranges, and any initial observations. Write this summary to a notebook or text file as the data dictionary.

Step 4: Data Cleaning

This step follows a deliberate prompt, plan, agree, then execute workflow. Do not ask AI to "just clean the data." Instead:

- First, ask: "What is the percentage of missing values in each column?"
- Then ask: "Are there any anomalies, outliers, or impossible values?"
- Then ask: "Based on what you found, propose a cleaning plan."
- Review the plan, agree or modify, then execute.

This teaches the discipline of understanding the data before touching it.

Step 5: Feature Engineering

Review the business questions. If any require derived features, create them here. For example, if Popularity needs to be encoded as ordinal (Very Low, Low, Moderate, High, Very High) rather than nominal, that decision is made at this stage. Keep it minimal. Only engineer what the business questions demand.

Step 6: Exploratory Data Analysis

EDA is done in three stages, all grounded to the business questions.

Univariate Analysis: Examine each variable individually. Histograms and box plots for numerical features. Bar charts for categorical features. The goal is to understand distributions, spot skewness, and identify outliers. Each plot should tie back to a business question.

Bivariate Analysis: Examine relationships between each predictor and Sales. Scatter plots for numerical predictors vs. Sales. Box plots for categorical predictors vs. Sales. Compute Pearson's correlation coefficient for each numerical predictor vs. Sales.

Multivariate Analysis: Build a correlation heatmap of all numerical variables. Identify which predictors might be redundant (highly correlated with each other) and which are independently useful. This sets up the modeling stage.

Step 7: Prepare Dataset for Modeling

Encode categorical variables using dummy or one-hot encoding. Perform a train-test split (for example, 70-30 or 80-20). No scaling is needed for linear regression. Document the split sizes and rationale.

Step 8: Model Building

Simple Linear Regression: Build individual models, one predictor at a time against Sales. Start with Advertising Expenditure, then Campaign Engagement Score, Discount Percentage, and so on. For each, interpret the coefficient and plot the best-fit line. Record R^2 for each single-predictor model.

Multiple Linear Regression: Combine all significant predictors (including encoded categorical variables) into one model. Interpret the coefficients, where each now represents the effect of that predictor holding all others constant. Compare the multiple regression R^2 to the individual simple regression R^2 values.

Step 9: Evaluation

Evaluate the model using R^2 , adjusted R^2 , RMSE, and MAE on the test set. Compare train vs. test performance to check for overfitting. Check regression assumptions:

- Linearity: Residuals vs. fitted values plot. Should be a random scatter with no pattern.
- Normality of residuals: Histogram of residuals and Q-Q plot.
- Homoscedasticity: Residuals vs. fitted values plot. Spread should be roughly constant, no funnel shape.
- Multicollinearity: Use the correlation heatmap from EDA. Flag any predictor pairs above 0.7 to 0.8.

Step 10: Business Recommendations

Go back to the business_questions.txt file. Answer each question using EDA findings and model results. Rank the predictors by coefficient magnitude to answer "which lever matters most." Translate into 3 to 5 plain-language recommendations for the marketing and merchandising teams.

Risk, Mitigation and Evaluation Criteria

Risks and Mitigations

Stakeholders read coefficients as causal guarantees (People): The marketing team may interpret "ad spend coefficient = \$28 per dollar" as a guaranteed outcome. To mitigate this,

frame all findings as "associated with" rather than "causes." Include a clear note in the final presentation. Recommend that any major budget shift be piloted in one region first before a full rollout.

Linear model misses non-linear patterns (Process): If discounts have diminishing returns after 30%, or if ad spend has a ceiling effect, a straight-line model will not capture that. To mitigate this, visually inspect scatter plots for curves during bivariate EDA. If found, document them as known limitations and flag as future scope for more advanced modeling.

Multicollinearity inflates coefficients (Process): If two predictors are highly correlated, the individual coefficients become unreliable. To mitigate this, use the correlation heatmap from EDA to visually identify any pairs above 0.7 to 0.8. If found, consider dropping one or noting the limitation. Since VIF is out of scope, the heatmap serves as the primary check.

Evaluation Criteria

- R^2 of at least 0.50 on the test set. Adjusted R^2 should be close to R^2 , indicating no overfitting from too many dummy variables.
- RMSE and MAE on the test set should be meaningfully lower than the standard deviation of Sales (roughly \$3,612).
- Residual vs. fitted plot shows no clear pattern. Q-Q plot residuals fall roughly along the diagonal. Correlation heatmap shows no predictor pairs above 0.8.
- Every question in business_questions.txt is answered with data. At least 3 actionable recommendations are delivered with supporting coefficients.
- Model RMSE compared to a naive baseline (predicting mean sales for every product) to show the model does meaningfully better.

Expected Outcomes and Impact

Short Term (0 to 4 Weeks)

A validated regression model that quantifies the relationship between each lever and sales. Based on the data's correlation structure, Advertising Expenditure is expected to emerge as the strongest driver (correlation of roughly 0.64), followed by Campaign Engagement Score (roughly 0.39) and Discount Percentage (roughly 0.25). The model will confirm whether Product Price and Return Rate have the expected negative association, and whether Length of Product Description can be safely deprioritized given its near-zero correlation with sales. Most importantly, the business questions file will be fully answered, giving the marketing team a

data-backed answer to "where should we put our next dollar?" and the merchandising team a better demand estimate than flat historical averages.

Long Term (1 to 6 Months)

If the regression insights are acted on, for example by reallocating 15% of marketing budget from low-impact areas to advertising and campaign engagement, ElectraHub could reasonably expect a measurable lift in sales over the next one to two quarters. The structured AI-assisted workflow (project setup, business questions, data understanding, cleaning, EDA, model, recommendations) becomes a repeatable template the analytics team can reuse for future datasets and business problems. The EDA and regression groundwork also set the stage for future modeling. Once the team is comfortable with linear regression, they can explore non-linear models, interaction effects, or regularization techniques as a natural next step.

Why Now

The online electronics market is intensely competitive. ElectraHub's competitors are already using data-driven inventory and marketing optimization. Every quarter spent making gut-based decisions across 3,000+ product-campaign cycles means potentially significant budget misallocation. The data already exists, is clean with no missing values, and is ready to analyze. The cost of this project is essentially 4 weeks of analysis. The cost of not doing it is continued guesswork on a multi-million-dollar marketing and inventory operation.

Cost of Inaction

Without understanding which levers actually drive sales, ElectraHub continues spreading marketing budget evenly, overinvesting in factors that do not move the needle and underinvesting in those that do. The 10x range in sales across the dataset (\$3,258 to \$34,421) shows there is massive variance to explain. A simple linear regression model will not capture everything, but it gives them a structured, quantified foundation they currently lack entirely. Every quarter of delay is a quarter of decisions made on gut feel while competitors use data.